

# VIJAYA KARTHIK

Generative AI Engineer | Production Multimodal RAG | Agentic AI | MLOps & Responsible AI

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## PROFESSIONAL SUMMARY

Generative AI Engineer who designed, built and shipped a production multimodal Retrieval-Augmented Generation (RAG) platform on AWS, spanning hybrid vector retrieval, agentic tool calling, self-hosted quantized LLM inference and a 12-module guardrail layer. Backed by 2,247 automated tests and a 30-metric MLflow-versioned evaluation harness whose CI gates block any merge that regresses retrieval or hallucination safety.

## TECHNICAL SKILLS

**Generative AI & LLMs:** RAG, Agentic AI, Tool Calling, Prompt Engineering, LLM Inference Optimization, llama.cpp, GGUF Quantization, Hugging Face Transformers, LangChain

**Retrieval & Vector Search:** Vector Databases, Qdrant, Hybrid Search (BM25 + Dense), Cross-Encoder Reranking, BGE Embeddings & Reranker, Maximal Marginal Relevance (MMR), Semantic Chunking

**Evaluation, Safety & Responsible AI:** RAGAS, DeepEval, MLflow, LLM-as-a-Judge, Hallucination Detection, Red-Teaming, Prompt Injection Defence, Presidio (PII), OWASP LLM Top 10

**ML, NLP & Multimodal:** PyTorch, TensorFlow / Keras, scikit-learn, CUDA, GPU Inference, Vision-Language Models (Qwen2-VL), Whisper ASR, OCR (Tesseract, EasyOCR), OpenCV

**Backend & Data:** Python, FastAPI, REST APIs, Pydantic v2, Async Python (asyncio), MongoDB, Redis, Pandas, NumPy, JWT, OAuth 2.0 (PKCE), Multi-Tenant Isolation

**Cloud, MLOps & Observability:** AWS (EC2 GPU, Lambda, CloudWatch, IAM, SSM), Docker, GitHub Actions CI/CD, Prometheus, Grafana, OpenTelemetry, pytest, Git, Linux

## PROJECTS

### Multimodal Agentic RAG Integrated Knowledge AI Assistant (MAGIK) v1.0.0 | Mar 2026 - Aug 2026

Self-hosted, fully open-source stack | Finance domain

github.com/vjkarthik98/MULTIMODAL-AGENTIC-RAG-INTEGRATED-KNOWLEDGE-AI-ASSISTANT

- **Multimodal ingestion:** Engineered seven isolated ingestion pipelines (text, PDF, DOCX, XLSX, image, audio, video), each owning its extraction, chunking, embedding and BM25 layer so a defect in one cannot degrade another.
- **Hybrid retrieval:** Fused BM25 sparse search with dense vector search on Qdrant, re-ranked by a BGE cross-encoder and diversified with MMR; 0.14s median / 0.23s p95 latency, CI-gated on a 56-query gold set.
- **Agentic routing:** Implemented an agentic query router that classifies each query (RAG, web search, hybrid, direct, finance sub-domain) and dispatches it through a typed ToolCall / ToolResult contract, held to a 0.917 CI accuracy floor.
- **Answer verification:** Grounded every claim and figure against retrieved evidence with a self-verifying layer that retries when checks fail; 1.00 citation accuracy and 0.93 grounding success measured across all seven modalities.
- **Guardrails & multi-tenancy:** Hardened all 28 pipeline surfaces with a 12-module guardrail layer (prompt injection, jailbreak, Presidio PII redaction, SSRF, rate limiting) against a 109-prompt red-team corpus - 100% injection recall (64/64) at 0.9% false positives - with tenant data isolated across four storage layers behind JWT, OAuth 2.0 (PKCE) and TOTP MFA.
- **GPU inference:** Ran 17 open-source models (~25 GB of weights) concurrently on one 48 GB NVIDIA L40S GPU under an explicit VRAM budget, serving a quantized Qwen2.5-14B GGUF model via llama.cpp - no third-party LLM API in the request path.
- **AWS deployment & CI/CD:** Deployed and operated the platform on AWS (EC2 GPU, Docker Compose, Caddy TLS, SSM secrets) across 9 GitHub Actions workflows with zero-downtime releases, auto-rollback and Lambda-driven GPU scale-to-zero.
- **Evaluation & observability:** Built the RAG evaluation harness (RAGAS, DeepEval, LLM-as-a-Judge) versioned in MLflow by Git SHA and prompt version, with CI gates holding fabrication rate at or below 0.072; traced with Prometheus, Grafana and OpenTelemetry.

## EXPERIENCE

### Generative AI Engineer (Self-Directed Practice) | Jun 2025 - Present

- **Jun 2025 - Mar 2026:** Full-time foundation in Python, machine learning, deep learning, transformer architectures, NLP, MLOps and Responsible AI - completed to three professional certificates and carried straight into production work.
- **Mar 2026 - Aug 2026:** Delivered MAGIK to v1.0.0 in under six months across 42 tagged releases - trunk-based Git workflow, pre-commit quality and security gates (Ruff, mypy, Bandit, pip-audit), and CI that blocks any merge on test, security or evaluation regressions.

## CERTIFICATIONS

- **IBM** - Generative AI Engineering Professional Certificate, 2025
- **Duke University** - Machine Learning Operations (MLOps) Specialization, 2026
- **Google Cloud** - Responsible AI for Developers, 2026

## EDUCATION

UPSC Civil Services Examination - Full-time preparation | 2021 - 2025

Master of Arts (English) - Bishop Heber College, Tiruchirappalli | 2019 - 2021

Bachelor of Arts (English) - Don Bosco College, Dharmapuri | 2016 - 2019